微信Mai_ange

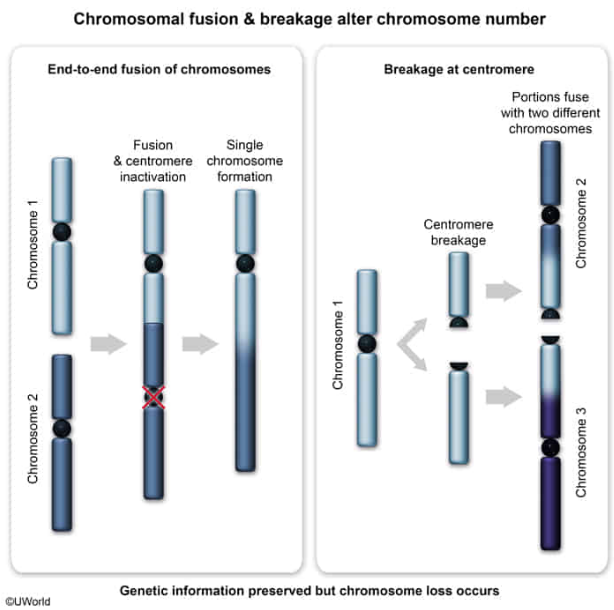
Scientists studying evolution in yeast found that a certain species underwent a reduction in chromosome number with no loss of coding information. Given this, which of the following mechanisms best explains how this reduction in chromosome number may have occurred?

I. End-to-end fusion of two chromosomes followed by inactivation of one centromere on the newly formed chromosome
II. Breakage of a chromosome at the centromere and fusion of the two resulting chromosomal portions to the ends of other chromosomes
III. Transfer of a DNA sequence from one chromosomal arm to a different chromosomal arm

- A. I only [9%]
- B. II only [15%]
- ✔ C. I and II only [57%]
- D. I and III only [18%]

Correct
55% answered correctly

Explanation:



Eukaryotic genomes consist of linear chromosomes made up of DNA that can be translated and transcribed into proteins. Each chromosome has a condensed region of repetitive noncoding DNA called the centromere, a structure that facilitates spindle fiber attachment during mitosis and meiosis. In addition, eukaryotic chromosomal ends consist of repetitive noncoding sequences called **telomeres** that protect coding information from degradation during DNA replication.

微信Mai_ange

Alterations to chromosome structure (eg, fusion or breakage) can occur due to DNA mutations, and chromosomal alterations may cause changes in chromosome number within a particular species. In this scenario, scientists studying evolution in yeast, a **single-celled eukaryotic organism**, discovered a species that underwent a reduction in chromosome number without losing any coding information. Accordingly, this reduction in chromosome number may have occurred via:

- **end-to-end** (telomere-to-telomere) **fusion of two chromosomes and inactivation of one of the centromeres**. This fusion would initially generate a larger chromosome with two centromeres, and inactivation of one of these centromeres would produce a *single* new chromosome, reducing the chromosome number in the cell by one **(Number I)**.

- **breakage of a chromosome at the centromere and fusion of each chromosomal portion to the ends of other chromosomes**. This initial breakage would result in two individual chromosomal portions. Fusion of these portions to other chromosomes would cause the original chromosome to be lost, also reducing the overall chromosome number in the cell by one **(Number II)**.

As both chromosomal events involve the *transfer* of *intact* genetic material to different chromosomes, neither would lead to a loss of coding information. In addition, because telomeres and centromeres are noncoding sequences, changes to these sequences due to chromosomal fusion or breakage would *not* likely affect protein-coding information.

(Number III) Transposons are small fragments of DNA that can move between different regions of the genome. Movement of a DNA sequence, such as a transposon, from one chromosomal arm to another may *alter* the nucleotide sequence of each chromosome but would *not* reduce chromosome number in the yeast cell.

Educational objective:
Eukaryotic chromosomes have both protein-coding DNA and distinct noncoding regions. Noncoding regions include the centromere (facilitates spindle fiber attachment during cell division) and telomeres (protect chromosomal ends from degradation).

Time spent:0 sec | **QID**:402600
Subject: Biology | **System:** Cellular Biology

微信Mai_ange